

scalars in the other. Show that the induced map $q^*: H^*(\mathbb{CP}^\infty; \mathbb{Z}) \rightarrow H^*(\mathbb{RP}^\infty; \mathbb{Z})$ is surjective in even dimensions by showing first by a geometric argument that the restriction $q: \mathbb{RP}^2 \rightarrow \mathbb{CP}^1$ induces a surjection on H^2 and then appealing to cup product structures. Next, form a quotient space X of $\mathbb{RP}^\infty \amalg \mathbb{CP}^n$ by identifying each point $x \in \mathbb{RP}^{2n}$ with $q(x) \in \mathbb{CP}^n$. Show there are ring isomorphisms $H^*(X; \mathbb{Z}) \approx \mathbb{Z}[\alpha]/(2\alpha^{n+1})$ and $H^*(X; \mathbb{Z}_2) \approx \mathbb{Z}_2[\alpha, \beta]/(\beta^2 - \alpha^{2n+1})$, where $|\alpha| = 2$ and $|\beta| = 2n + 1$. Make a similar construction and analysis for the quotient map $q: \mathbb{CP}^\infty \rightarrow \mathbb{HP}^\infty$.

15. For a fixed coefficient field F , define the **Poincaré series** of a space X to be the formal power series $p(t) = \sum_i a_i t^i$ where a_i is the dimension of $H^i(X; F)$ as a vector space over F , assuming this dimension is finite for all i . Show that $p(X \times Y) = p(X)p(Y)$. Compute the Poincaré series for S^n , $\mathbb{RP}^n$, $\mathbb{RP}^\infty$, $\mathbb{CP}^n$, $\mathbb{CP}^\infty$, and the spaces in the preceding three exercises.

16. Show that if X and Y are finite CW complexes such that $H^*(X; \mathbb{Z})$ and $H^*(Y; \mathbb{Z})$ contain no elements of order a power of a given prime p , then the same is true for $X \times Y$. [Apply Theorem 3.16 with coefficients in various fields.]

17. Show that $H^*(J(S^n); \mathbb{Z})$ for n odd is isomorphic to $H^*(S^n; \mathbb{Z}) \otimes H^*(J(S^{2n}); \mathbb{Z})$ as a graded ring. [Consider the natural quotient map $S^n \times S^n \times J_{2k-1}(S^n) \rightarrow J_{2k+1}(S^n)$ and use induction on k .]

18. For the closed orientable surface M of genus $g \geq 1$, show that for each nonzero $\alpha \in H^1(M; \mathbb{Z})$ there exists $\beta \in H^1(M; \mathbb{Z})$ with $\alpha\beta \neq 0$. Deduce that M is not homotopy equivalent to a wedge sum $X \vee Y$ of CW complexes with nontrivial reduced homology. Do the same for closed nonorientable surfaces using cohomology with $\mathbb{Z}_2$ coefficients.

3.3 Poincaré Duality

Algebraic topology is most often concerned with properties of spaces that depend only on homotopy type, so local topological properties do not play much of a role. Digressing somewhat from this viewpoint, we study in this section a class of spaces whose most prominent feature is their local topology, namely manifolds, which are locally homeomorphic to $\mathbb{R}^n$. It is somewhat miraculous that just this local homogeneity property, together with global compactness, is enough to impose a strong symmetry on the homology and cohomology groups of such spaces, as well as strong nontriviality of cup products. This is the Poincaré duality theorem, one of the earliest theorems in the subject. In fact, Poincaré's original work on the duality property came before homology and cohomology had even been properly defined, and it took many

years for the concepts of homology and cohomology to be refined sufficiently to put Poincaré duality on a firm footing.

Let us begin with some definitions. A **manifold of dimension n** , or more concisely an **n -manifold**, is a Hausdorff space M in which each point has an open neighborhood homeomorphic to $\mathbb{R}^n$. The dimension of M is intrinsically characterized by the fact that for $x \in M$, the local homology group $H_i(M, M - \{x\}; \mathbb{Z})$ is nonzero only for $i = n$:

$$\begin{aligned} H_i(M, M - \{x\}; \mathbb{Z}) &\approx H_i(\mathbb{R}^n, \mathbb{R}^n - \{0\}; \mathbb{Z}) && \text{by excision} \\ &\approx \tilde{H}_{i-1}(\mathbb{R}^n - \{0\}; \mathbb{Z}) && \text{since } \mathbb{R}^n \text{ is contractible} \\ &\approx \tilde{H}_{i-1}(S^{n-1}; \mathbb{Z}) && \text{since } \mathbb{R}^n - \{0\} \simeq S^{n-1} \end{aligned}$$

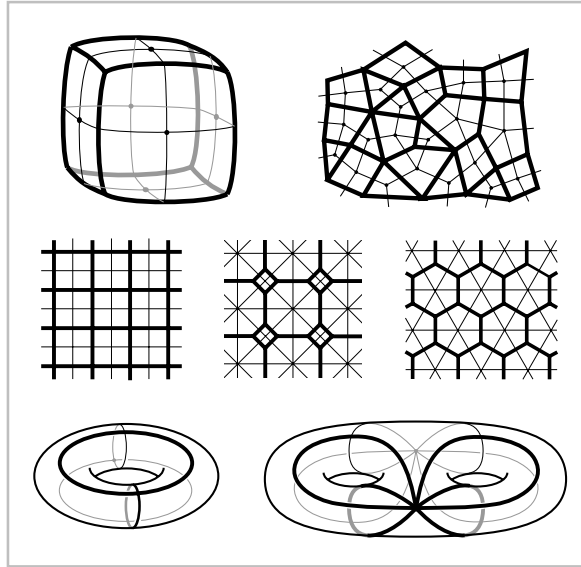
A compact manifold is called **closed**, to distinguish it from the more general notion of a compact manifold with boundary, considered later in this section. For example S^n is a closed manifold, as are $\mathbb{R}P^n$ and lens spaces since they have S^n as a covering space. Another closed manifold is $\mathbb{C}P^n$. This is compact since it is a quotient space of S^{2n+1} , and the manifold property is satisfied since there is an open cover by subsets homeomorphic to $\mathbb{R}^{2n}$, the sets $U_i = \{[z_0, \dots, z_n] \in \mathbb{C}P^n \mid z_i = 1\}$. The same reasoning applies also for quaternionic projective spaces. Further examples of closed manifolds can be generated from these using the obvious fact that the product of closed manifolds of dimensions m and n is a closed manifold of dimension $m + n$.

Poincaré duality in its most primitive form asserts that for a closed orientable manifold M of dimension n , there are isomorphisms $H_k(M; \mathbb{Z}) \approx H^{n-k}(M; \mathbb{Z})$ for all k . Implicit here is the convention that homology and cohomology groups of negative dimension are zero, so the duality statement includes the fact that all the non-trivial homology and cohomology of M lies in the dimension range from 0 to n . The definition of ‘orientable’ will be given below. Without the orientability hypothesis there is a weaker statement that $H_k(M; \mathbb{Z}_2) \approx H^{n-k}(M; \mathbb{Z}_2)$ for all k . As we show in Corollaries A.8 and A.9 in the Appendix, the homology groups of a closed manifold are all finitely generated. So via the universal coefficient theorem, Poincaré duality for a closed orientable n -manifold M can be stated just in terms of homology: Modulo their torsion subgroups, $H_k(M; \mathbb{Z})$ and $H_{n-k}(M; \mathbb{Z})$ are isomorphic, and the torsion subgroups of $H_k(M; \mathbb{Z})$ and $H_{n-k-1}(M; \mathbb{Z})$ are isomorphic. However, the statement in terms of cohomology is really more natural.

Poincaré duality thus expresses a certain symmetry in the homology of closed orientable manifolds. For example, consider the n -dimensional torus T^n , the product of n circles. By induction on n it follows from the Künneth formula, or from the easy special case $H_i(X \times S^1; \mathbb{Z}) \approx H_i(X; \mathbb{Z}) \oplus H_{i-1}(X; \mathbb{Z})$ which was an exercise in §2.2, that $H_k(T^n; \mathbb{Z})$ is isomorphic to the direct sum of $\binom{n}{k}$ copies of $\mathbb{Z}$. So Poincaré duality is reflected in the relation $\binom{n}{k} = \binom{n}{n-k}$. The reader might also check that Poincaré duality is consistent with our calculations of the homology of projective spaces and lens spaces, which are all orientable except for $\mathbb{R}P^n$ with n even.

For many manifolds there is a very nice geometric proof of Poincaré duality using the notion of *dual cell structures*. The germ of this idea can be traced back to the five regular Platonic solids: the tetrahedron, cube, octahedron, dodecahedron, and icosahedron. Each of these polyhedra has a dual polyhedron whose vertices are the center points of the faces of the given polyhedron. Thus the dual of the cube is the octahedron, and vice versa. Similarly the dodecahedron and icosahedron are dual to each other, and the tetrahedron is its own dual. One can regard each of these polyhedra as defining a cell structure C on S^2 with a dual cell structure C^* determined by the dual polyhedron. Each vertex of C lies in a dual 2-cell of C^* , each edge of C crosses a dual edge of C^* , and each 2-cell of C contains a dual vertex of C^* .

The first figure at the right shows the case of the cube and octahedron. There is no need to restrict to regular polyhedra here, and we can generalize further by replacing S^2 by any surface. A portion of a more-or-less random pair of dual cell structures is shown in the second figure. On the torus, if we lift a dual pair of cell structures to the universal cover $\mathbb{R}^2$, we get a dual pair of periodic tilings of the plane, as in the next three figures. The last two figures show that the standard CW structure on the surface of genus g , obtained from a $4g$ -gon by identifying edges via the product of commutators $[a_1, b_1] \cdots [a_g, b_g]$, is homeomorphic to its own dual.



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Given a pair of dual cell structures C and C^* on a closed surface M , the pairing of cells with dual cells gives identifications of cellular chain groups $C_0^* = C_2$, $C_1^* = C_1$, and $C_2^* = C_0$. If we use $\mathbb{Z}$ coefficients these identifications are not quite canonical since there is an ambiguity of sign for each cell, the choice of a generator for the corresponding $\mathbb{Z}$ summand of the cellular chain complex. We can avoid this ambiguity by considering the simpler situation of $\mathbb{Z}_2$ coefficients, where the identifications $C_i = C_{2-i}^*$ are completely canonical. The key observation now is that under these identifications, the cellular boundary map $\partial: C_i \rightarrow C_{i-1}$ becomes the cellular coboundary map $\delta: C_{2-i}^* \rightarrow C_{2-i+1}^*$ since ∂ assigns to a cell the sum of the cells which are faces of it, while δ assigns to a cell the sum of the cells of which it is a face. Thus $H_i(C; \mathbb{Z}_2) \approx H^{2-i}(C^*; \mathbb{Z}_2)$, and hence $H_i(M; \mathbb{Z}_2) \approx H^{2-i}(M; \mathbb{Z}_2)$ since C and C^* are cell structures on the same surface M .

To refine this argument to $\mathbb{Z}$ coefficients the problem of signs must be addressed. After analyzing the situation more closely, one sees that if M is orientable, it is possible to make consistent choices of orientations of all the cells of C and C^* so that the boundary maps in C agree with the coboundary maps in C^* , and therefore one gets $H_i(C; \mathbb{Z}) \approx H^{2-i}(C^*; \mathbb{Z})$, hence $H_i(M; \mathbb{Z}) \approx H^{2-i}(M; \mathbb{Z})$.

For manifolds of higher dimension the situation is entirely analogous. One would consider dual cell structures C and C^* on a closed n -manifold M , each i -cell of C being dual to a unique $(n-i)$ -cell of C^* which it intersects in one point ‘transversely.’ For example on the 3-dimensional torus $S^1 \times S^1 \times S^1$ one could take the standard cell structure lifting to the decomposition of the universal cover $\mathbb{R}^3$ into cubes with vertices at the integer lattice points $\mathbb{Z}^3$, and then the dual cell structure is obtained by translating this by the vector $(\frac{1}{2}, \frac{1}{2}, \frac{1}{2})$. Each edge in either cell structure then has a dual 2-cell which it pierces orthogonally, and each vertex lies in a dual 3-cell.

All the manifolds one commonly meets, for example all differentiable manifolds, have dually paired cell structures with the properties needed to carry out the proof of Poincaré duality we have just sketched. However, to construct these cell structures requires a certain amount of manifold theory. To avoid this, and to get a theorem that applies to all manifolds, we will take a completely different approach, using algebraic topology to replace the geometry of dual cell structures.

Orientations and Homology

Let us consider the question of how one might define orientability for manifolds. First there is the local question: What is an orientation of $\mathbb{R}^n$? Whatever an orientation of $\mathbb{R}^n$ is, it should have the property that it is preserved under rotations and reversed by reflections. For example, in $\mathbb{R}^2$ the notions of ‘clockwise’ and ‘counterclockwise’ certainly have this property, as do ‘right-handed’ and ‘left-handed’ in $\mathbb{R}^3$. We shall take the viewpoint that this property is what characterizes orientations, so anything satisfying the property can be regarded as an orientation.

With this in mind, we propose the following as an algebraic-topological definition: An orientation of $\mathbb{R}^n$ at a point x is a choice of generator of the infinite cyclic group $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\})$, where the absence of a coefficient group from the notation means that we take coefficients in $\mathbb{Z}$. To verify that the characteristic property of orientations is satisfied we use the isomorphisms $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \approx H_{n-1}(\mathbb{R}^n - \{x\}) \approx H_{n-1}(S^{n-1})$ where S^{n-1} is a sphere centered at x . Since these isomorphisms are natural, and rotations of S^{n-1} have degree 1, being homotopic to the identity, while reflections have degree -1 , we see that a rotation ρ of $\mathbb{R}^n$ fixing x takes a generator α of $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\})$ to itself, $\rho_*(\alpha) = \alpha$, while a reflection takes α to $-\alpha$.

Note that with this definition, an orientation of $\mathbb{R}^n$ at a point x determines an orientation at every other point y via the canonical isomorphisms $H_n(\mathbb{R}^n, \mathbb{R}^n - \{x\}) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - B) \approx H_n(\mathbb{R}^n, \mathbb{R}^n - \{y\})$ where B is any ball containing both x and y .